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Chapter 3

Requirements Analysis and System Investigation

3.1 Introduction

To thoroughly understand the RUET PoC (Point of Contact) app system and identify its underlying architecture, requirements, and operational challenges, the most critical step is effective information gathering. This process involves using various tools known as information gathering tools, which play a vital role in system analysis.

In our case, information was collected in multiple stages:

1. Initial data was obtained from publicly available sources such as the RUET ICT Cell's official documentation, system specifications, and published technical reports. These provided insights into the system's objectives, technical architecture, and organizational context within RUET.
2. Direct data collection was carried out through structured interviews and questionnaires with key officials and technical personnel from the RUET ICT Cell. This approach allowed us to gather in-depth knowledge about the system's day-to-day operations, internal workflows, technical infrastructure, current capabilities, and future requirements.
3. On-site observation and system documentation review provided practical insights into actual system usage, deployment environment, and real-world operational challenges.

By combining both documentary resources and primary data collection techniques, we achieved a comprehensive view of the RUET PoC system under study.

3.2 Information Gathering Tools

To collect information from the RUET ICT Cell regarding the PoC app system, we adopted a structured and systematic approach. Our team visited the ICT Cell to directly observe the system infrastructure and operational environment. Additionally, we engaged in direct communication with the Director, Senior System Analysts, and technical personnel to gain deeper insights.

The information collection process was carried out in four distinct steps:

3.2.1 On-Site Observation

Our team observed the RUET ICT Cell operations and infrastructure, which allowed us to study how the system is deployed, maintained, and used in real-time.

3.2.2 Structured Interviews

We conducted structured interviews with key officials including the Director, Senior System Analysts, Assistant Computer Programmer, and Network Engineer to gather qualitative insights into system architecture, operational processes, current challenges, technical capabilities, and perspectives on the system's effectiveness and future requirements.

3.2.3 Questionnaires

Pre-designed questionnaires were used to collect quantitative and standardized information from the technical and administrative staff, enabling us to compare responses, identify patterns, and understand user satisfaction and requirements from multiple stakeholders.

This multi-step approach ensured that the information gathered was both comprehensive and reliable, supporting our understanding of the system from multiple perspectives.

3.3 Review of Literature, Procedure and System Documentation

Our team gathered essential information from the RUET ICT Cell's official documentation, system specifications, and technical reports, which provided a clearer understanding of the organization's background, technical structure, and system operations.

3.3.1 RUET ICT Cell Background and Organization

The RUET ICT Cell was established to provide comprehensive information technology infrastructure and digital service delivery across Rajshahi University of Engineering & Technology. The cell operates under the administrative structure of RUET and is responsible for managing institutional digital systems, supporting academic operations, and enabling inter-institutional communication.

3.3.2 Overview of RUET PoC System

The RUET PoC (Point of Contact) app is a centralized directory and contact management system designed to consolidate staff and faculty information across the university. The system serves as a baseline automation initiative, providing a single point of access for institutional contact information.

System Architecture Overview

1. **Backend Infrastructure:** Core PHP application server with MySQL relational database, deployed on BdRen C-panel managed hosting server.
2. **Database Design:** Single centralized SQL database handling all services, ensuring data consistency and eliminating historical ambiguities in employee identification and email assignment.
3. **Frontend Interfaces:** Android mobile application for end-user access, web portal at poc.ruet.ac.bd for administrative functions, and JavaScript for client-side interactivity.
4. **User Base:** Primary focus on Teachers (faculty members) and Officers (administrative staff), with future expansion planned to include Students and low-level support staff.
5. **Data Scope:** Employee directory information including name, designation, department, employee ID, contact details (mobile, IP phone, office mail, personal mail), office location hierarchy, joining date, and 300x300 photograph.

Collected System Information

Technical Stack:

1. Current Version: Core PHP, MySQL, JavaScript
2. Planned Version 2: Laravel framework, MySQL, jQuery, Tailwind CSS
3. Deployment: BdRen C-panel hosting server

4. Mobile Platform: Android primary, iOS unreleased
5. Online Dependency: Currently requires full online connectivity

Organizational Team Structure:

1. Director: Dr. Md. Samiul Habib (Overall ICT supervision and strategic planning)
2. Senior System Analysts: S Mustafa Kamal, Engr. Shihan Arafin (System analysis and technical architecture)
3. Assistant Computer Programmer: Engr. Md. Julfikar Islam (Backend development)
4. Assistant Network Engineer: Md Ashif Al Amin (Infrastructure and hosting)
5. Lab Assistant: Md. Shahnewaz Uddin (Technical support and testing)

3.4 On-Site Observation

During our visit to the RUET ICT Cell, we were genuinely impressed by the overall technical environment and professional approach to system development. Upon arrival, we first approached the administrative office and introduced ourselves, explaining the purpose of our visit related to academic coursework on system analysis.

We were directed to meet with the Senior System Analysts and technical team members. They warmly welcomed us and expressed their willingness to support our study of the PoC system.

3.4.1 Workplace Environment and Operations

While observing the ICT Cell operations, we observed the following:

1. The technical team appeared highly engaged in system maintenance and development activities, demonstrating strong collaboration and technical expertise.
2. The server infrastructure was well-maintained with proper documentation and access control measures in place to ensure data security and system reliability.
3. The team demonstrated coordinated effort in managing both current system operations and planning for Version 2 upgrades, showing both reactive maintenance and proactive improvement initiatives.

4. The working environment was professional and collaborative, with clear communication channels between administrators, developers, network engineers, and support staff.
5. Documentation practices appeared comprehensive, with technical specifications, database schemas, and system procedures well-organized and accessible to authorized personnel.

3.4.2 System Infrastructure Assessment

1. Server infrastructure was observed to be stable and properly configured on BdRen C-panel hosting platform.
2. Database systems (MySQL) were observed to maintain consistent data integrity with proper backup and access control procedures.
3. Administrative interfaces and web portal (poc.ruet.ac.bd) were functional and provided comprehensive system management capabilities.
4. Mobile app deployment was observed to be actively used by designated staff members for directory access and contact information retrieval.

3.5 Structured Interviews

When we explained our academic research purpose to the team, the Senior System Analysts appointed Engr. Shihan Arafin to conduct a structured technical interview with us, providing detailed insights into the system's architecture, operations, and requirements.

3.5.1 Interview with Senior System Analyst

System Architecture and Technical Foundation

Q1: Can you provide us with a comprehensive overview of the RUET PoC app's technical architecture and system design? The RUET PoC app is built on a centralized, relational database architecture using MySQL with a Core PHP backend server. The system follows a three-tier architecture:

1. **Presentation Layer:** Android mobile application for end-users and web-based administrative portal at poc.ruet.ac.bd.
2. **Application Layer:** Core PHP backend processing user requests, managing authentication, and handling business logic.

3. **Data Layer:** Centralized MySQL database storing all staff information, categorized into Teacher, Officer, and Staff modules.

The entire system is deployed on BdRen C-panel managed hosting infrastructure, ensuring reliable uptime and professional server management.

Q2: What data elements are currently stored in the RUET PoC system? The system maintains the following comprehensive data fields for each staff member:

1. Name and Employee ID (unique institutional identifier)
2. Designation and Department assignment
3. First Joining Date (employment start date)
4. Contact Information: Mobile Number, IP Phone, Office Mail, Personal Mail
5. Office Location: Building, Floor, Room hierarchy for precise physical localization
6. Photograph: 300x300 pixel image for visual identification and verification
7. Module Classification: Teacher, Officer, or Staff categorization

Q3: What is the data input process and workflow for populating the directory? Data entry is performed through the administrative web interface at poc.ruet.ac.bd by authorized ICT Cell personnel. The workflow follows these steps:

1. Manual data entry of staff information through standardized web forms
2. Validation and verification of entered information for accuracy
3. Categorization of staff into appropriate modules (Teacher, Officer, Staff)
4. Storage in the centralized SQL database
5. Synchronization and automatic distribution to the mobile app
6. Users access current directory information through the Android app or web portal

The system currently operates in full online mode, requiring continuous network connectivity for all functionality.

System Components and Technology Stack

Q4: Can you describe the complete technology stack used in the RUET PoC application? The current production system uses the following technology stack:

1. **Backend Framework:** Core PHP (procedural PHP without modern framework)
2. **Database:** MySQL with relational schema design
3. **Frontend (Web):** HTML/CSS with vanilla JavaScript for client-side interactivity
4. **Frontend (Mobile):** Android native application
5. **Server:** BdRen C-panel managed hosting
6. **Version Control:** Git-based repository management

For Version 2, significant technology upgrades are planned:

1. **Backend Framework:** Migration to Laravel for improved modularity and maintainability
2. **Frontend:** jQuery for enhanced client-side interactivity and Tailwind CSS for modern UI/UX design
3. **Database:** MySQL maintained with optimized schema design
4. **Architecture:** Planning for offline-first capabilities with potential Firebase synchronization

Q5: What is the current user base and scope of the RUET PoC app? The system currently serves three primary user categories:

1. **Teachers (Faculty Members):** Academic faculty and teaching staff with complete contact and location information.
2. **Officers (Administrative Staff):** Administrative and support officers managing various university functions.
3. **System Administrators:** ICT Cell personnel managing data entry, user access, and system maintenance.

User categories currently **not included** in the system:

1. Students (excluded from current scope, planned for future expansion)
2. Third and fourth class employees (low-level manual workers) - challenging to integrate due to employment structure and contact information inconsistencies

Operational Challenges and System Limitations

Q6: What are the primary technical challenges currently faced by the RUET PoC system? The system faces several significant challenges:

1. **Inclusion Gap:** The current system structure effectively serves Teachers and Officers but struggles to cleanly incorporate low-level employees (3rd/4th class workers) due to irregular employment patterns and inconsistent contact information.
2. **Online Dependency:** The system requires continuous network connectivity with no offline caching capability. This creates fragility in areas with poor network coverage and limits accessibility.
3. **Data Quality Issues:** Legacy photographs uploaded during initial deployment lack standardized verification. Manual QA processes for photo verification create bottlenecks as the system scales.
4. **Technology Stack Limitations:** Core PHP lacks modularity and maintainability features required for modern application development. The current stack makes it difficult to add new features or scale the system.

Q7: How are data accuracy and currency currently maintained in the system? Data accuracy is maintained through the following mechanisms:

1. Periodic manual verification by ICT Cell administrators
2. Direct data entry by authorized personnel with access controls
3. Email notifications to staff members requesting data updates when organizational changes occur
4. Manual review process before data goes live in the directory

However, challenges remain:

1. Delays in reflecting organizational changes from the source systems
2. Inconsistencies in photograph verification and standards
3. Limited automated validation of contact information accuracy
4. Manual processes become inefficient as the user base grows

Future Development and Enhancement Plans

Q8: What are the planned enhancements and future roadmap for the RUET PoC system? The RUET PoC system has a structured development roadmap:

Short-term Improvements (Version 1.x):

1. Add Search functionality with advanced filtering capabilities
2. Implement Add to Favorites feature for quick access to frequently accessed contacts
3. Improve user interface design and user experience
4. Enhance error handling and system reliability

Medium-term Upgrades (Version 2):

1. Complete technology stack migration to Laravel, jQuery, and Tailwind CSS
2. Implementation of offline-first architecture with local data caching
3. Firebase or alternative cloud synchronization for hybrid online/offline functionality
4. Database schema optimization and improved indexing for performance

Long-term Expansion:

1. Integration of additional institutional modules (Guest House Booking System, Postgraduate Application System)
2. Facial recognition implementation for automated photo verification
3. Student directory integration and services
4. Mobile app expansion to iOS with feature parity to Android version
5. Enhanced search and recommendation features using machine learning

Q9: How will the system address the challenge of including low-level employees? Addressing the inclusion gap requires multiple approaches:

1. **Simplified Registration Flow:** Development of streamlined data entry process for employees without regular office assignments.

2. **Phone-Based Verification:** Implementation of alternative verification methods beyond email for employees with limited institutional email usage.
3. **Flexible Data Schema:** Modifications to system to accommodate variable employment arrangements and contact information patterns.
4. **Mobile-First Approach:** Enhanced mobile app capabilities for self-service profile management and updates.

3.6 Questionnaire

To better understand system requirements, technical priorities, and satisfaction levels among technical staff, our team developed a questionnaire focusing on various aspects of system functionality, technical adequacy, and development priorities. The survey was administered to 8 technical and administrative personnel using confidential forms, each with multiple-choice options to ensure anonymity and encourage honest feedback.

Below is a question-wise breakdown with response options and summarized findings.

3.6.1 Q1: How satisfied are you with the current RUET PoC app's technical architecture?

Options:

- a. Not Satisfied
- b. Less Satisfied
- c. Maybe Satisfied
- d. Satisfied
- e. Very Satisfied

Findings:

1. 37.5% selected d. Satisfied
2. 37.5% selected c. Maybe Satisfied
3. 25% selected b. Less Satisfied

Interpretation: While the current system is functional, there is recognition that the architecture has limitations. Satisfaction levels indicate readiness for technology stack upgrades to modern frameworks.

3.6.2 Q2: How well does the current system meet the information storage and retrieval requirements?

Options:

- a. Very Poorly
- b. Poorly
- c. Adequately
- d. Well
- e. Very Well

Findings:

- 1. 50% selected d. Well
- 2. 37.5% selected c. Adequately
- 3. 12.5% selected b. Poorly

Interpretation: The system successfully stores and retrieves directory information, meeting basic functional requirements. However, some gaps exist in advanced retrieval functionality.

3.6.3 Q3: What is your priority for the next system enhancement?

Options:

- a. Offline-first functionality
- b. Technology stack upgrade (Laravel/Tailwind)
- c. Search and filter improvements
- d. Enhanced user interface
- e. System security enhancements

Findings:

1. 37.5% selected b. Technology stack upgrade
2. 25% selected a. Offline-first functionality
3. 25% selected c. Search and filter improvements
4. 12.5% selected d. Enhanced user interface

Interpretation: Technical staff prioritizes modernization of the technology stack, recognizing the importance of using contemporary frameworks for maintainability and future scalability. Offline functionality and improved search are also significant priorities.

3.6.4 Q4: How adequately does the current system handle the inclusion of diverse employee categories?

Options:

- a. Very Inadequately
- b. Inadequately
- c. Somewhat Adequately
- d. Adequately
- e. Very Adequately

Findings:

1. 50% selected b. Inadequately
2. 37.5% selected c. Somewhat Adequately
3. 12.5% selected d. Adequately

Interpretation: There is consensus that the system needs improvement in handling diverse employee categories, particularly low-level staff. This confirms the identified inclusion gap challenge.

3.6.5 Q5: Are you satisfied with the current MySQL database design and data structure?

Options:

- a. Not Satisfied
- b. Less Satisfied
- c. Maybe Satisfied
- d. Satisfied
- e. Very Satisfied

Findings:

- 1. 50% selected d. Satisfied
- 2. 37.5% selected c. Maybe Satisfied
- 3. 12.5% selected a. Not Satisfied

Interpretation: The relational database design effectively supports current requirements. However, some recognition exists that future scaling and feature additions may require schema optimization.

3.6.6 Q6: How well does the current system handle data security and confidentiality?

Options:

- a. Very Poorly
- b. Poorly
- c. Adequately
- d. Well
- e. Very Well

Findings:

1. 62.5% selected d. Well
2. 25% selected c. Adequately
3. 12.5% selected b. Poorly

Interpretation: Data security is generally well-maintained through current practices. However, some concerns exist regarding potential security enhancements as the system expands.

3.6.7 Q7: What is your overall confidence in the system's ability to scale to future requirements?

Options:

- a. Very Low
- b. Low
- c. Moderate
- d. High
- e. Very High

Findings:

1. 37.5% selected c. Moderate
2. 37.5% selected d. High
3. 25% selected b. Low

Interpretation: While the system has demonstrated foundational capability, confidence in long-term scalability is moderate. The planned Version 2 upgrade is critical to improving scalability and maintainability for future growth.

3.7 Conclusion

This comprehensive investigation of the RUET PoC app system, conducted through systematic documentation review, on-site observation, structured interviews, and staff questionnaires, has yielded significant insights into the system's current state, capabilities, and future requirements.

3.7.1 Summary of Findings

The RUET PoC app represents a successful initial automation effort at RUET, establishing a centralized, authoritative source for staff directory information. The single-database SQL design ensures data consistency, eliminates historical ambiguities in employee identification and email assignment, and provides reliable operational functionality for Teachers and Officers.

3.7.2 System Strengths

1. Stable and reliable centralized database architecture
2. Clear organizational team structure with defined responsibilities
3. Well-maintained server infrastructure on professional hosting platform
4. Effective data storage and retrieval mechanisms for current user base
5. Professional approach to system maintenance and development
6. Clear development roadmap and modernization strategy

3.7.3 Key System Characteristics

1. **Technology Stack:** Core PHP (current), planned migration to Laravel/Tailwind (Version 2)
2. **Data Architecture:** Centralized MySQL database with relational design
3. **User Base:** Teachers and Officers primarily, with planned expansion to Students
4. **Operational Mode:** Full online dependency in current version, offline capability planned for Version 2
5. **Data Scope:** Comprehensive contact and location information for all categorized staff members

3.7.4 Key Requirements Identified

1. Technology stack modernization to Laravel/Tailwind/jQuery framework stack
2. Implementation of offline-first architecture with Firebase synchronization
3. Development of mechanisms for low-level employee inclusion
4. Enhancement of search and filtering capabilities

5. Implementation of automated photo verification using facial recognition
6. Integration of future institutional modules and services
7. User interface improvements and enhanced user experience

3.7.5 Strategic Recommendations

The investigation indicates that the RUET PoC system is well-positioned for continued development and enhancement. Priority should be given to:

1. Accelerating Version 2 technology stack migration to improve maintainability and scalability
2. Implementing offline-first functionality to enhance system reliability in diverse network environments
3. Developing inclusive mechanisms for low-level employee integration
4. Establishing automated data quality processes to reduce manual verification overhead
5. Planning for modular architecture to support future institutional module integration

Through systematic implementation of these requirements and recommendations, the RUET PoC app can evolve into a robust, scalable platform serving the complete institutional communication and directory needs of RUET.